

	Authentication	<i>[Define authentication method for users] Login via OTP using Fast2SMS for Owner, Staff, Customer</i>	High
1	Authorization	<i>[Define role-based access control] Role based: Owner - full access; Staff - order management & inventory; Customer - place order & view loyalty points</i>	High
3	External interfaces	<i>[Detail third-party APIs, external databases, or services the system must connect to] Razorpay Payment Gateway, Fast2SMS SMS API, Google Maps for delivery tracking, WhatsApp Business API</i>	High
4	Transactions processing	<i>[Explain how data exchanges, payments, or core system transactions will be handled] Chai orders billing, inventory deduction, loyalty points redemption, UPI/cash payment processing</i>	High
5	Reporting	<i>[Describe the types of reports, dashboards, or data exports the system must generate] Sales reports daily/monthly, customer analytics, inventory reports dashboard with charts</i>	Medium
6	Business rules, etc.	<i>[List specific logic, constraints, or workflows unique to the business process] Auto loyalty: 1 point per Rs10 spent; stock alert below 5kg; 10% discount on 5th order; referral bonus for new customer referral</i>	High
7	Compliance to laws or regulations	<i>[Specify any legal, industry, or data privacy standards the system must adhere to (e.g., GDPR, HIPAA)] Data privacy for mobile numbers, secure storage, GDPR consent for SMS communications</i>	Medium
8	Other	<i>[Add any additional functional requirements here] Customer feedback collection & ratings, referral rewards program, WhatsApp notification for offer alerts</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	System responds under 2 sec for billing & search under 1 sec	Target: <2 seconds page load
2	Scalability	Handles 500 concurrent users during peak rush; scalable for future growth	Supports 10,000 concurrent users
3	Security & Data Privacy	Encrypted data, mobile OTP authentication, role-based access, secure payment handling	AES-256 encryption
4	Reliability & Availability	99.5% uptime with daily backup and failover mechanism	99.9% uptime per month
5	Usability & Accessibility	Simple Tamil/English UI, mobile responsive design, 2-click ordering process	WCAG 2.1 AA standard
6	Maintainability	Easy to maintain using Docker & Git, modular code, portable deployment	Maintainability & Portability

Project Initialization and Planning Phase

Date	01.10.2026
Team ID	6ab295b91266c775575a6dce
Project Name	ACRM APPLICATION FOR THE CHAI SHOP & CO
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for your customers' challenges. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A male applicant seeking a loan.	Secure a loan for the property	Married with no co-applicant income.	Self-employed with a good credit history.	Optimistic about loan approval.
PS-2	Chai Shop Owner	Manage customer orders & loyalty	Manual tracking difficult	No digital CRM	Stressed
PS-3	Chai Shop Staff	Track inventory & sales	Frequent stock-outs	No automated alerts	Frustrated
PS-4	Regular Customer	Get quick services & rewards	Long waiting & no rewards	No CRM application	Disappointed

Empathy Map

Date	01.10.2026
Team ID	6ab295b91266c775575a6dce
Project Name	ACRM APPLICATION FOR THE CHAI S_HOP
Maximum Marks	4 Marks

Empathy Map

Create an empathy map to understand your customer/user from four perspectives - what they Say, Think, Do, and Feel - in relation to the problem your project addresses. This helps the team build a customer-centric solution.

SAYS • Need quick billing • Remember regular customers • Loyalty points venum	THINKS • How to increase sales? • Customer data manage? • Stock track pannalum?
PERSONA NAME <i>Chai Shop Owner</i>	
DOES • Writes orders in notebook • Forgets customer preference • Checks stock visually	FEELS • Stressed during rush hours • Worried about losing customers • Excited for digital solution



Date: 01.10.2026 | Team ID: 6ab295b91266c775575a6dce | Project Name: CRM APPLICATION FOR THE CHAI SHOP & CO

Maximum Marks: 4 Marks | Team Size: 5 Members | Team Members: Sivaranjani E, Sivashakthi S, Snega B, Sudhesh P, Suganya E

S.No	NFR Category	Requirement Description	Example Target Metric
1	Performance & Speed	System responds under 2 sec for billing operations; search results under 1 sec; order processing < 3 sec during peak hours	(e.g., Page loads in < 2 seconds)
2	Scalability	Handles 500 concurrent users during morning/evening rush hour; scalable to 10,000 users for chain expansion; MongoDB sharding for data growth	(e.g., Supports up to 10,000 concurrent users)
3	Security & Data Privacy	Mobile number encryption AES-256; OTP authentication; role-based access control for Owner/Staff/Customer; secure Razorpay payment handling	(e.g., End-to-end AES-256 encryption)
4	Reliability & Availability	99.5% uptime with daily automated backup; failover mechanism; offline billing capability when internet is down	(e.g., 99.9% uptime per month)
5	Usability & Accessibility	Simple Tamil/English bilingual UI; mobile responsive design; 2-click chai ordering; easy for non-technical staff	(e.g., Complies with WCAG 2.1 AA standards)
6	Other	Easy to maintain using Docker & Git modular code; portable deployment on Render/Vercel/ AWS; low maintenance for chai shop owner	(e.g., Maintainability & Portability)

Brainstorming & Idea Prioritization Template

Date	01.10.2026
Team ID	6ab295b91266c775575a6dce
Project Name	ACRM APPLICATION FOR THE CHAI SHOP & CO
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Sivaraajani E	Digital Order Management System		
2	Sivasbakthi S	Loyalty Rewards	Customer Engagement	
3	Snega B	Inventory Tracking	Inventory	Group 3
4	Sudhesh P	Sales Analytics Dash	Analytics	Group 1
5	Suganya E	Feedback Collection	Feedback	Group 2
6	Team Combined	Automated Billing	Operations	Group 1

Step 2: Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

Group No.	Final Idea	Feasibility (High/Medium/Low)	Importance (High/Medium/Low)	Priority	Selected (Yes/No)
1	Order & Sales Management	High	High	Priority 1	Yes
2	Loyalty & Feedback Module	High	High	Priority 2	Yes
3	Inventory Alert & Stock	Medium	High	Priority 3	Yes

Data Flow Diagram

Date	01.10.2026
Team ID	6ab295b91266c77557a6de
Project Name	CRM APPLICATION FOR THE CHAI SHOP & CO
Maximum Marks	2 Marks

Data Flow Diagram

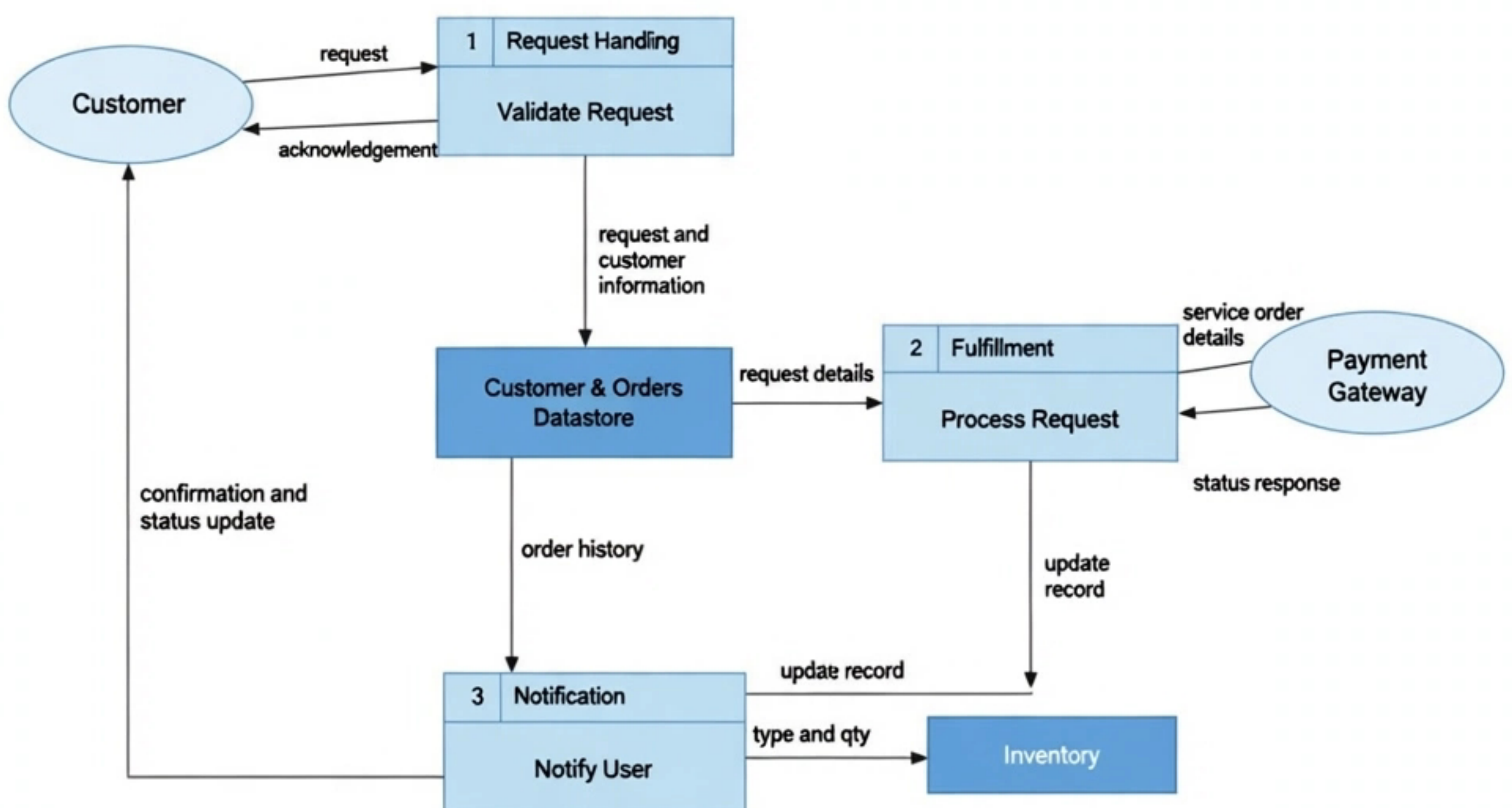
Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.



S.No	Architecture Component Layer	Technology Chosen	Justification / Purpose
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>MongoDB Atlas NoSQL Database with Mongoose ODM</i>	<i>Handles unstructured data like customer feedback & orders; flexible schema for chai inventory variations; strong scalability for growing customer base; cloud-hosted, no server maintenance</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Render for Backend Hosting + Vercel for Frontend + AWS S3 for Image Storage + Docker Containerization</i>	<i>Easy scalability during rush hours; cost-effective hosting for small chai shop startup; free tier available; portable deployment; reliable cloud infrastructure</i>
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	<i>Git + GitHub for Version Control + Render Auto Deploy CI/CD</i>	<i>Seamless collaboration among 5 team members; automated testing before deploy; easy rollback if billing issue; branch management for features: loyalty & payment</i>
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Razorpay Payment Gateway + Fast2SMS SMS API + WhatsApp Business API + Google Maps API + JWT + bcrypt</i>	<i>Enables UPI/cash secure payments; SMS alerts for order ready & loyalty points credit; WhatsApp offer notifications & referral discounts; delivery delivery tracking; authentication for Owner/Staff/Customer</i>



Customer Journey Map

Date	01.10.2026
Team ID	ACRM APPLICATION FOR THE CHAI SHOP & CO
Project Name	SWTID-2026-3164
Maximum Marks	2 Marks

Customer Journey Map

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

Phase of Journey	Stage 1	Stage 2	Stage 3
Actions <i>What does the customer do?</i>	• Customer enters shop • sees menu •	Place order Pay Wait	• Give feedback • Redeem rewards •
Touchpoint <i>What part of the service do they interact with?</i>	• • •	Counter POS • CRM App	• Feedback form • Loyalty app • <i>•Will come again?</i>
Customer Thought <i>What is the customer thinking?</i>	• • •	• Will my loyalty points add? • Excited	• • •
Customer Feeling <i>What is the customer feeling?</i>			<i>•Satisfied Loyal</i>



Process Ownership <i>Who is in the lead on this?</i>			
Opportunities <i>How can we improve this stage?</i>	Staff • Digital menu display	Cashier • Quick billing via CRM	Owner • Loyalty rewards & personalized offers